Personal Finance Analytics App
iOS Course Project Proposal

Student Information

Student: Guntash Brar
Course: iOS Application Development
Platform: iOS
Development Environment: Xcode
Programming Language: Swift

1. Project Overview

The Personal Finance Analytics App is an iOS application designed to help users track their daily expenses and gain insights into their spending habits. The application will allow users to record financial transactions, categorize expenses, and view visual summaries of their spending patterns.

Many people struggle to understand where their money is going each month. This application aims to simplify personal financial tracking by presenting data in a clear and visual format. The app will provide charts and summaries that help users analyze their financial behavior and make better financial decisions.

The application will be built using SwiftUI, Apple's modern framework for building user interfaces on iOS. Financial data will be stored locally on the device using CoreData or local storage, and spending insights will be displayed using Swift Charts.

2. Objectives

The primary objectives of this project are:

- Develop a functional iOS application using SwiftUI
- Implement a system for tracking personal expenses
- Store financial data locally on the device
- Use data visualization techniques to present financial insights
- Provide users with a simple interface to analyze their spending patterns

3. Features

Expense Tracking

Users will be able to add new expenses to the application. Each expense entry will include:
• amount spent
• category of expense
• date of transaction

Spending Categories

Expenses will be grouped into categories to help organize financial data. Example categories may include:
• Food
• Rent
• Transportation
• Entertainment
• Utilities
• Other

Data Visualization

The application will display financial data using visual charts such as:
• Pie Chart showing spending distribution across categories
• Monthly Spending Line Chart showing trends over time
• Budget Progress Bar showing how much of a monthly budget has been used

Budget Tracking

Users will be able to define a monthly budget. The application will track expenses against this budget and visually display progress toward the budget limit.

Spending Insights

The application will provide insights such as:
• highest spending category
• monthly spending trends
• budget usage percentage

4. Technologies Used

SwiftUI – Used to build the user interface.
CoreData / Local Storage – Used to store expense data locally on the device.
Swift Charts – Used to generate charts and visualizations for financial data.
Xcode – Development environment used to build and test the application.

5. System Design

The application will follow a Model-View architecture.

Model

The model will manage the financial data including:
• expense entries
• categories
• budgets

View

The view layer will handle the user interface and display:
• expense lists
• charts and analytics
• budget tracking information

Data Storage

Financial records will be saved locally on the device using CoreData or local storage so the application can retain user data between sessions.

6. Expected Outcome

The final application will allow users to easily record their expenses and visualize their financial habits through charts and summaries.

This project demonstrates several important iOS development concepts including:
• SwiftUI interface design
• local data storage
• data visualization
• user interaction and state management

The completed application will provide a useful personal finance tool while demonstrating practical mobile development skills.

7. Repository

The project repository will be publicly available on GitHub.

Repository Link:
https://github.com/guntash499/FinanceAnalyticsApp

The repository will contain the application source code, documentation, and project proposal.